

KARRATHA KING BAY, Australia

WMO: 953080

Lat: 20.617S

Lon: 116.750E

Elev: 16

StdP: 101.13

Time Zone: 8.00 (E08)

Period: 91-01

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.3	17.1	-3.2	2.9	23.7	-0.8	3.5	23.1	15.7	20.3	14.6	19.7	8.0	110	0.587

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	6.5	36.6	23.7	35.2	24.2	34.2	24.6	29.6	32.3	28.6	31.7	27.9	31.3	6.9	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.1	25.8	31.4	27.8	23.8	30.8	27.0	22.7	30.3	98.5	32.4	93.1	32.0	89.5	31.3	34.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.6	13.1	11.6	DB	13.5	41.4	2.0	2.5	12.0	43.2	10.8	44.7	9.7	46.1	8.2	47.9	(4)
(5)			WB	7.4	31.5	0.9	2.3	6.8	33.1	6.2	34.4	5.7	35.7	5.0	37.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.4	29.7	29.9	30.4	29.1	25.2	22.2	21.5	22.8	23.8	26.4	27.4	28.9	(6)	
	DBStd	3.70	1.74	1.86	1.90	1.75	2.09	2.32	1.67	1.82	1.81	2.42	2.31	2.11	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	3	0	0	0	0	0	1	2	1	0	0	0	0	(9)	
	CDD10.0	5997	609	558	633	574	472	367	357	396	415	508	521	586	(10)	
	CDD18.3	2958	351	325	375	324	214	118	101	137	165	250	271	327	(11)	
	CDH23.3	30469	4536	4316	5043	3928	1611	456	252	561	871	2210	2758	3928	(12)	
	CDH26.7	12592	2120	2127	2621	1729	374	48	11	72	151	714	972	1656	(13)	
(14) Wind	WSAvg	6.4	6.8	6.8	6.1	5.4	5.9	6.4	6.1	5.8	6.0	6.6	7.3	7.3	(14)	
(15) Precipitation	PrecAvg	300	50	86	43	20	31	37	11	2	1	1	2	12	(15)	
	PrecMax	756	248	359	206	124	163	165	70	23	19	14	39	106	(16)	
	PrecMin	69	0	0	0	0	1	0	1	0	0	0	0	0	(17)	
	PrecStd	182	68	105	60	31	40	46	16	6	4	3	8	26	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.3	37.8	38.5	36.1	32.9	30.1	28.0	30.2	32.7	36.5	36.6	38.1	(19)	
		MCWB	24.0	25.1	22.5	21.8	20.2	20.5	16.4	16.6	19.4	20.0	20.8	21.8	(20)	
	2%	DB	34.7	35.4	36.2	34.8	31.2	27.9	26.6	28.4	30.0	33.1	33.9	34.7	(21)	
		MCWB	26.8	26.7	23.6	21.7	20.8	19.6	16.8	16.5	19.9	20.3	21.8	25.3	(22)	
	5%	DB	33.6	34.1	34.8	33.7	29.9	26.8	25.6	27.3	28.4	31.7	32.3	33.3	(23)	
		MCWB	26.7	26.6	24.8	22.0	20.6	19.4	17.2	16.9	19.3	20.5	22.6	25.6	(24)	
	10%	DB	32.5	33.0	33.7	32.6	28.7	25.8	24.8	26.3	27.2	30.2	31.0	32.0	(25)	
		MCWB	26.6	26.6	25.2	22.5	20.7	19.2	17.1	17.1	19.3	21.0	22.7	25.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	31.0	30.8	30.8	29.6	25.7	24.2	23.5	22.7	24.6	25.6	28.0	29.8	(27)	
		MCDB	33.7	33.6	33.1	32.2	28.8	26.5	24.7	24.2	28.8	29.4	31.1	32.3	(28)	
	2%	WB	29.5	29.2	28.3	27.1	24.7	22.9	21.6	21.6	22.7	24.7	26.5	28.6	(29)	
		MCDB	31.9	32.3	32.4	31.2	27.6	25.2	23.9	24.2	26.8	28.3	29.6	31.3	(30)	
	5%	WB	28.5	28.2	27.6	26.1	23.9	22.2	20.4	20.9	21.9	24.0	25.6	27.7	(31)	
		MCDB	31.2	31.9	31.8	29.9	27.1	25.0	23.6	24.0	26.0	27.9	29.0	30.6	(32)	
	10%	WB	27.6	27.5	27.0	25.4	23.1	21.2	19.4	20.2	21.2	23.3	24.9	26.9	(33)	
		MCDB	30.8	31.3	31.1	29.5	26.6	24.6	22.9	23.8	25.3	27.5	28.4	30.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	5.0	5.2	6.5	7.0	6.1	5.7	6.5	7.0	6.7	6.8	6.3	5.3	(35)	
		MCDBR	6.3	7.0	7.9	8.4	7.2	6.5	7.4	8.1	8.1	8.9	8.4	7.2	(36)	
	5% WB	MCWBR	3.9	4.5	5.2	5.9	4.8	4.4	5.3	5.5	6.0	5.9	5.5	4.4	(37)	
		MCWBR	5.3	5.8	6.4	6.9	6.0	4.9	6.0	6.3	6.8	7.0	6.5	5.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.402	0.397	0.378	0.353	0.335	0.323	0.316	0.314	0.333	0.357	0.359	0.378	(40)	
		taud	2.517	2.546	2.589	2.615	2.600	2.622	2.615	2.582	2.528	2.487	2.519	2.539	(41)	
	Ebn at Noon		944	937	930	915	895	889	906	941	959	964	978	967	(42)	
		Edh at Noon	114	109	100	91	85	80	83	92	105	114	113	111	(43)	
(44) All-Sky Solar Radiation	RadAvg		7.46	7.15	6.38	5.56	4.64	4.21	4.57	5.64	6.66	7.54	8.03	8.02	(44)	
	RadStd		0.45	0.49	0.46	0.30	0.35	0.30	0.29	0.16	0.25	0.16	0.18	0.31	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)		+0.46	N/A	N/A	N/A	+0.48	+0.56	N/A	N/A	N/A	+158

Nomenclature: See separate page